COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-INF-014: Network Switching Standard

Document ID	TS-INF-014
Title	Network Switching Standard
Executive Sponsor	Network Services Managers
Issuing Authority	Office of Information Technology (OIT)
Applies to	OIT, state agencies as defined in C.R.S. 24-37.5-101(28), and vendors contracted by OIT or its stakeholders.
Tier	Standard
Version	3.0
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1. Purpose

This standard is issued under the authority of the Governor's Office of Information Technology (OIT) (C.R.S. 24-37.5, et seq.) and the Office of Enterprise Architecture (Rule 8 CCR 1501-7). It communicates the standards OIT has developed to provide enterprise standard configurations for network switches, describing the minimum requirements for network switches.

2. Scope

These standards apply to OIT, state agencies as defined in C.R.S. 24-37.5-101(28), and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Standards instituted for use by approval of the OIT Architecture Review Board (ARB) govern purchase and deployment decisions deemed "go-forward" or "net-new", and are not intended as a retroactively enforceable mandate for immediate conformance, especially where cost implications may be prohibitive.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.

- Reviews and approves the Standard (this document).

Network Services Managers

- Owner of Standard; engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number; presents to the Architecture Review Board for approval and adoption.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Due to the variety of office sizes, building layouts, and networking requirements, a spectrum of common switching architectures is outlined. As agencies and OIT upgrade their network, it is crucial to procure devices that meet the minimum functionality required for seamless integration with the Colorado State Network (CSN) and Software-Defined Wide Area Network (SD-WAN).

Must-Have Access-Layer and Distribution Features

- Mandatory Core Features: Spanning Tree Protocol (STP), Gigabit capabilities, remote administration, SNMP, and MIB; PoE+ and Universal PoE (UPOE) models; high-capacity stacking; uplink options for Gigabit, 10 Gigabit, and 40 Gigabit Ethernet.
- Advanced Routing Features for Layer 3 Switches: IPv4 and IPv6 routing, static routes, and advanced IP routing protocols (OSPF, BGP, IS-IS); policy-based routing and multicast capabilities.
- Quality of Service (QoS) Features: Hierarchical bandwidth management, CoS, DSCP, shaping, and policing.
- Mandatory Security Features: IEEE 802.1x, DHCP snooping, dynamic ARP inspection, RA guard, IP source guard, and ACLs; management via SSHv2 and SNMPv3; TACACS+ and RADIUS support; RSA key generation of at least 2048 bits.
- Maintenance and Monitoring: Port monitoring capabilities; SolarWinds ability to read SNMP MIBs; centralized management platform connectivity; enhanced warranties with next business day (NBD) advance hardware replacement.

Site Size Recommendations

- Small Sites (0–45 endpoints): Gigabit (PoE) ports, 1 Gig uplink, advanced spanning tree protocols.
- Medium Sites (50–100+ endpoints): Similar features with 10 Gig uplinks.
- Large Locations (150+ endpoints): Switch stacks connected to a distribution switch with L3 capabilities.

Installation Standards

- Mount the device in a 19" network rack (wall-mounted or floor standing) in a secured room with ventilation.
- Location must have power availability compatible with a power filter, UPS, or surge protection.
- Physical Security: The agency is responsible for securing the room and/or key management; switches must be installed in secure locations limited to authorized personnel only.
- Configuration management must follow CISP-005.

Refresh (End of Life)

- Remote Location Access Switches: 4–6 years recommended refresh cycle.
- Distribution Switches: 5–7 years, ideally updated with the full site refresh.
- Core/Chassis Switches: 7–8 years recommended refresh cycle.

5. Exceptions and Variance

Requests for variances from any provision of this standard must be requested using the Technical Standard End User Exception Request Procedure. Requests must include compensating controls. The Secure Configuration Exception Request form must be submitted to oit_ciso@state.co.us.

6. References

- OIT Hardware Standard Guidance
- Cisco Configuration Management: Best Practices
- Cisco Network Security Baseline
- Cisco Guide to Harden IOS Devices

7. Definitions

Access Layer: The lowest layer of the TCP/IP protocol, providing means for the system to deliver data to other devices directly attached to a network.

Network Switch: A computer networking device that connects devices together using packet switching to receive, process, and forward data.

Port Density: A switch with a large number of switch ports.

RADIUS: A networking protocol providing centralized AAA management for users connecting to a network service.

Redundancy: System design in which a component is duplicated so a backup exists if it fails.

TACACS+: A network protocol for remote authentication and authorization of users attempting to access network resources.

SolarWinds: IT monitoring and management tool for an enterprise environment.